Relative interaction Index on survival data

First approaches

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1 Results summary

Although the initial methodology used to calculate the RII generated a binomial distribution that is incompatible with standard parametric tests (ANOVA and t-tests), the results provide significant insights—specifically, the higher stress sensitivity of *Q. faginea* compared to *Q. ilex*. Consequently, alternative analytical approaches are currently being examined.

2 Introduction

Plant–plant interactions are key drivers of community composition and diversity. These interactions can be direct (between two species) or indirect (when a third species modifies the interaction between the first two) (Michalet et al. 2014).

- **Types of indirect effects:**
  - **Indirect facilitation:** Occurs when one species reduces the competition experienced by two other species Michalet et al. (2017).
  - **Indirect competition:** Occurs when a facilitator increases the biomass of a competing species to such an extent that it ultimately harms a third species (the original beneficiary) Michalet et al. (2017).
- **Interaction mechanisms:**
  - **Resource competition (exploitation):** Indirect interaction through a shared limiting resource (water, nutrients) (Michalet et al. 2017).
  - **Interference:** A direct form of competition in which one organism actively interferes with another's ability to acquire resources, for example through litter accumulation that acts as a physical barrier or alters soil moisture (Michalet et al. 2017).
  - **Canopy vs. root effects:** Plants may exert positive (thermal buffering) or negative (light competition) effects through their canopy, and variable effects through their roots (nutrient competition or facilitation via mycorrhizal associations) (Wang et al. 2017).
- **"Hidden" interactions:** At the community level, net effects may appear neutral when positive and negative interactions compensate each other, although these interactions still structure species composition Wang et al. (2017).

The use of the **RII (Relative Interaction Index)** responds to the need for a standardized comparative metric that quantifies both the **strength** and **direction** of biotic interactions between species. Unlike other indices, RII is **symmetric around zero**, enabling a balanced statistical interpretation where positive values indicate facilitation and negative values indicate competition, constrained between $+1$ and -1. This property is essential in complex community studies, as it allows the decomposition and mathematical comparison of different interaction components—such as canopy vs. root effects, or direct vs. indirect effects—revealing **"hidden interactions"** that may cancel out at the net community level Wang et al. (2017).

3 Objectives

1. To establish the methodology for applying RII calculations to our case study.
2. To explore the results of RII calculations.
3. To define the statistical methodology required to determine which RII-derived effects are significant.

4 Methodology

4.1 Experimental design

The study combined three main factors to analyze facilitation:

- **Environment (2 levels):** Two adjacent areas in Monte Público Jócar (Guadalajara, Spain) were selected: a *Pinus pinaster* forest (tree cover) and gap dominated by shrubland following a wildfire in 2014.
- **Microsite (4 levels):** Defined according to nurse shrub species: *Cistus ladanifer*, *Genista scorpius*, *Rosa canina*, and a vegetation-free control ("open area").
- **Facilitated species (2 levels):** Two oak species with contrasting ecological strategies were planted: *Quercus ilex* (evergreen and stress-tolerant) and *Quercus faginea* (deciduous and more drought-sensitive).

Sixty replicates per microsite were randomly selected, evenly distributed between the two environments (30 in pine forest and 30 in gap habitat), totaling 240 microsites. In each microsite, two individuals were planted: one from each *Quercus* species.

4.2 RII calculation

The RII is a standardized index used to quantify the strength and direction of biotic interactions. Its general formula is:

$$RII = \frac{X_{\text{with neighbors}} - X_{\text{without neighbors}}}{X_{\text{with neighbors}} + X_{\text{without neighbors}}}$$

Where X represents a measure of plant performance (commonly biomass, species richness, or abundance).

Properties of RII:

- Symmetric around zero
- Ranges from -1 (maximum competition) to $+1$ (maximum facilitation)

RII is typically calculated on continuous (biomass) or positive discrete data. However, its application to binary survival data (0/1) is less common, though still possible with adaptations.

RII in binary data (0/1)

RII can be computed in a similar way for binary data. For a pair of observations i and j (RII_{ij}), survival under shrub presence (S_{a_i}) is compared with survival in open conditions (S_{b_j}). Since values are binary, the result can only be $+1$, 0, or -1. When both S_{a_i} and S_{b_j} are 0, RII_{ij} is defined as 0 by convention.

$$RII_{ij} = \frac{S_{a_i} - S_{b_j}}{S_{a_i} + S_{b_j}}$$

{#eq-RII}

except when:

$$S_a + S_B = 0;;\Rightarrow;;RII = 0$$

Table 1: Possible RII outcomes for binary (0/1) data

(S_{a_i})	(S_{b_j})	(RII_{ij})
1	0	1
0	1	-1
1	1	0
0	0	0 (by convention)

RII in our experimental design

Comparison against all controls

As shown in Section 4.1, the design does not include paired controls. This means that each survival observation under a shrub does not have a directly associated survival observation in a nearby shrub-free location. Therefore, for any given shrub-associated observation, there is no single predefined control replicate to use for RII calculation. This lack of pairing is a consequence of the original experimental design, which was not initially developed with RII computation in mind.

Since no control is uniquely matched to each observation, a brute-force approach was used: survival under each shrub individual is compared against survival from all control plots. Given that there are 30 control replicates, each oak seedling under each shrub yields 30 RII values (**?@eq-RII**).

For example, to compute the RII for the direct interaction under *Cistus ladanifer* individual 1 (RII_{Cistus_1}), 30 RII values are computed, one for each control replicate ($RII_{Cistus_1-Control_1}, \dots, RII_{Cistus_1-Control_{30}}$

A single value of RII_{Cistus_1} is then obtained by averaging all pairwise values:

$$\overline{RII}_{Cistus_1} = \frac{1}{m}\sum_{j=1}^{m} RII_{Cistus_1 j}$$

Defining meaningful comparisons

The computation of a single RII per individual requires defining which comparisons are biologically meaningful for the research questions.

We impose the following constraints:

1. All RII values are computed within the same oak species. For example, it is not meaningful to compare *Quercus ilex* under *Cistus ladanifer* with *Quercus faginea* under the same shrub. The experimental design assumes that the planted oaks are young enough (one year old) and sufficiently spaced to avoid direct competition.
2. **Direct effects of shrubs** (RII^{shrubs}_{direct}): Effect of shrubs on oak survival without pine influence, comparing shrub microsites vs. open area in the gap environment.
3. **Indirect effects of shrubs** ($RII^{shrubs}_{indirect}$): Effect of shrubs on oak survival under pine cover, comparing shrub microsites vs. without-shrubs within the pine forest.
4. **Direct effects of pine** (RII^{pine}_{direct}): Effect of pine forest on oak survival in open area (no shrubs), comparing pine vs. gap conditions.
5. **Indirect effects of pine** ($RII^{pine}_{indirect}$): Effect of pine on oak survival under shrubs, comparing pine vs. open conditions within shrub microsites. This would capture contextual effects of pine mediated by shrub presence. However, this is not central to the study's theoretical assumptions, which treat pine as a background environmental modifier rather than a target of shrub feedbacks.

Effect	Type	Comparison (treatment vs control)	Pine condition	Shrub condition	Interpretation
RII^{shrubs}_{direct}	Shrubs (direct)	Under shrub vs open	Without pine	Shrub vs open	Direct shrub effect on oak survival without pine influence
$RII^{shrubs}_{indirect}$	Shrubs (indirect)	Under shrub vs open	With pine	Shrub vs open	Shrub effect on oak survival in presence of pine
RII^{pine}_{direct}	Pine (direct)	Pine vs no pine (gap)	In open (no shrub)	No shrub	Direct pine effect on survival without shrubs

4.3 Statistical analysis

To determine whether the detected effects are statistically significant and how they influence the community, two methods will be used:

- **One-sample t-test:** This test is used to assess whether the mean RII value is significantly different from zero. If it is, a significant effect is present (facilitative if > 0, competitive if < 0).
- **ANOVA:** This is used to compare whether different types of RII (e.g., direct vs. indirect RII) differ significantly from each other, indicating a significant indirect effect.

5 Results

5.1 Description

```
library(tidyverse)


source("01-scripts/01.21-survival.R")

# 1. Prepare data ----

# Select variables of interest
df.surv <- df.surv |>
  select(Individual, Environment, microsite, quercus_sp, survival, surv_date)
  ↪ |>
  mutate(survival = if_else(survival == 2, 1, survival))

# Create subdatasets per quercus species
df.surv.qf <- df.surv |>
  filter(quercus_sp == "QF")

df.surv.qi <- df.surv |>
  filter(quercus_sp == "QI")

# 2. Describe comparisons ----
# Each row: focal group (A) vs reference group (B).

comparisons <- tibble(
  env_A  = c("pine canopy", "pine canopy",    "pine canopy",
```

```
             "gap",    "gap",        "gap",
             "pine canopy"),
  ms_A     = c("Cistus ladanifer",  "Rosa canina",    "Genista scorpius",
             "Cistus ladanifer",  "Rosa canina",    "Genista scorpius",
             "open"),
  env_B   = c("pine canopy", "pine canopy",    "pine canopy",
             "gap",    "gap",        "gap",
             "gap"),
  ms_B     = c("open",          "open",             "open",
             "open",          "open",             "open",
             "open"),
  type_RII = c("Indirect", "Indirect", "Indirect",
                "Direct",    "Direct",    "Direct",
              "Direct"),
  Interacting_species = c("Cistus ladanifer", "Rosa canina", "Genista
   ↪ scorpius",
                          "Cistus ladanifer", "Rosa canina", "Genista
                           ↪ scorpius",
                          "Pinus pinaster")
)

# 3. Compute RII ----
## Create function ----
calc_rii <- function(df, env_A, ms_A, env_B, ms_B, type_RII,
 ↪ Interacting_species) {

  # Create focal group
  focal <- df |>
    filter(Environment == env_A, microsite == ms_A) |>
    select(
      ind_A = Individual,
      env_A = Environment,
      ms_A = microsite,
      surv_A = survival,
      surv_date
    )

  # Create reference group
  ref <- df |>
    filter(Environment == env_B, microsite == ms_B) |>
    select(
```

```
      ind_B = Individual,
      env_B = Environment,
      ms_B = microsite,
      surv_B = survival,
      surv_date
    )

  # Inner join allow to make comparison only within the same date
  inner_join(focal, ref, by = "surv_date") |>
    mutate(
      type_RII = type_RII,
      Interacting_species = Interacting_species,
      RII = if_else(
        surv_A + surv_B == 0,
        0,
        (surv_A - surv_B) / (surv_A + surv_B)
      )
    )
}

## Apply function to every comparison and combine ----

df.rii.qi <- pmap(comparisons, calc_rii, df = df.surv.qi) |>
  bind_rows()

df.rii.qf <- pmap(comparisons, calc_rii, df = df.surv.qf) |>
  bind_rows()

## Compute aggregated RII per individual

aggID.df.rii.qi <- df.rii.qi |>
  group_by(ind_A, env_A, ms_A, surv_date, type_RII, Interacting_species) |>
  summarise(
    mean = mean(RII, na.rm = TRUE),
    sd   = sd(RII, na.rm = TRUE)
  )

aggID.df.rii.qf <- df.rii.qf |>
  group_by(ind_A, env_A, ms_A, surv_date, type_RII, Interacting_species) |>
  summarise(
    mean = mean(RII, na.rm = TRUE),
```

```
    sd   = sd(RII, na.rm = TRUE)
  )
```

Holm oak (Quercus ilex)

In the direct shrub effects, values are positive for all three nurse species at both sampling dates. This indicates that, in the absence of pine forest, *Q. ilex* survival was generally higher under shrubs than in gaps. The strongest effect appears in the direct effect of *Pinus pinaster*. Across dates, the facilitative effect of *Cistus ladanifer* becomes stronger in May 2026, while the effect of *Genista scorpius* moves closer to zero (Figure 1).

In the indirect shrub effects, the pattern shifts slightly. *Cistus ladanifer* shows negative values at both dates (−0.0805 and −0.145), suggesting that under pine cover its effect on *Q. ilex* survival is slightly less favorable than in gaps within the same environment. This is not a strong effect, but it does suggest a potential contextual inversion. The other two shrub species maintain positive effects, although the effect of *Rosa canina* becomes less clear in the second date (Figure 1).

Portuguese oak (Quercus faginea)

In *Q. faginea*, a more contrasted pattern emerges than in *Q. ilex*. In the direct effects, pine shows the highest RII values in both dates (0.567 and 0.530), indicating a clear and consistent facilitative effect of the pine canopy on survival relative to gap conditions. Among shrubs, *Cistus ladanifer* shows positive but more moderate values (0.233 and 0.133), while *Genista scorpius* and *Rosa canina* show weak direct effects, close to zero in the second date. This suggests that, for **Q. faginea*, the favourable microsite effect of shrubs is modest, whereas pine exerts a stronger influence (Figure 1).

In the indirect effects, the most striking pattern appears: *Cistus ladanifer* shows clearly negative values at both dates (−0.287 and −0.330), substantially more negative than in *Q. ilex*. This indicates that, under pine cover, the presence of *Cistus* is associated with a reduction in *Q. faginea* survival relative to open (without shrubs) within the same environment. In contrast, *Genista scorpius* and *Rosa canina* remain close to zero (0.0333–0.0370 and 0.0667–0.00370), suggesting very weak or nearly neutral indirect effects. Overall, *Q. faginea* appears more sensitive to the contextual conditions imposed by pine and by certain shrubs, especially *Cistus*, consistent with its higher sensitivity to environmental stress (Figure 1).

```
agg.rii.qi <-
aggID.df.rii.qi |>
  filter_out(surv_date == "12/06/2025") |>
  group_by(surv_date, type_RII, Interacting_species) |>
  summarise(
    n = sum(!is.na(mean)),
    media = mean(mean, na.rm = TRUE),
    se = sd(mean, na.rm = TRUE) / sqrt(n),
```

```r
    .groups = "drop"
  ) |>
  mutate(quercus_sp = "Q. ilex")

agg.rii.qf <-
  aggID.df.rii.qf |>
  filter_out(surv_date == "12/06/2025") |>
  group_by(surv_date, type_RII, Interacting_species) |>
  summarise(
    n = sum(!is.na(mean)),
    media = mean(mean, na.rm = TRUE),
    se = sd(mean, na.rm = TRUE) / sqrt(n),
    .groups = "drop"
  ) |>
  mutate(quercus_sp = "Q. faginea")

agg.rii <- rbind(agg.rii.qi, agg.rii.qf)

ggplot(agg.rii, aes(x = media, y = Interacting_species, colour = quercus_sp))
↪   +
  geom_point(position = position_dodge(width = .5), size = 1.5) +
  geom_errorbarh(aes(xmin = media - se, xmax = media + se), width = .5,
   ↪  position = position_dodge(width = .5)) +
  geom_vline(aes(xintercept = 0), linetype = "dashed", colour = "red") +
  facet_grid(surv_date~type_RII) +
  ylab("") +
  xlab("RII") +
  labs(title = "Direct and indirect RII") +
  coord_cartesian(xlim = c(-.6, .6)) +
  theme_light() +
  theme(legend.position = "bottom")
```

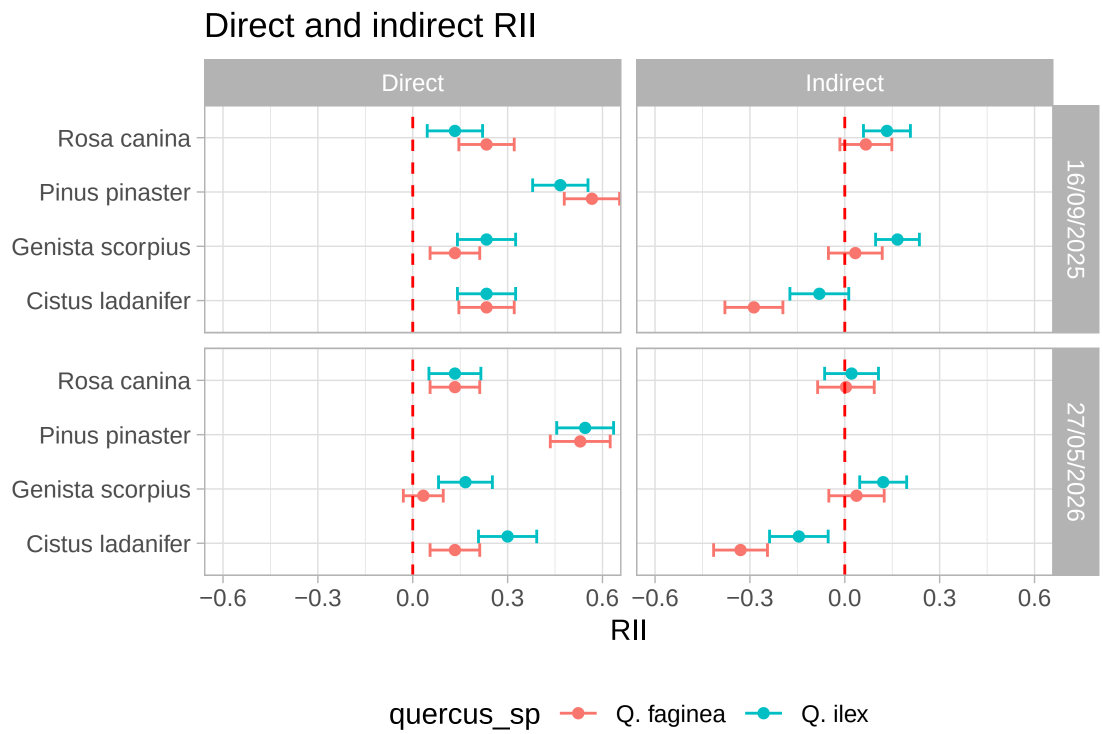


Figure 1

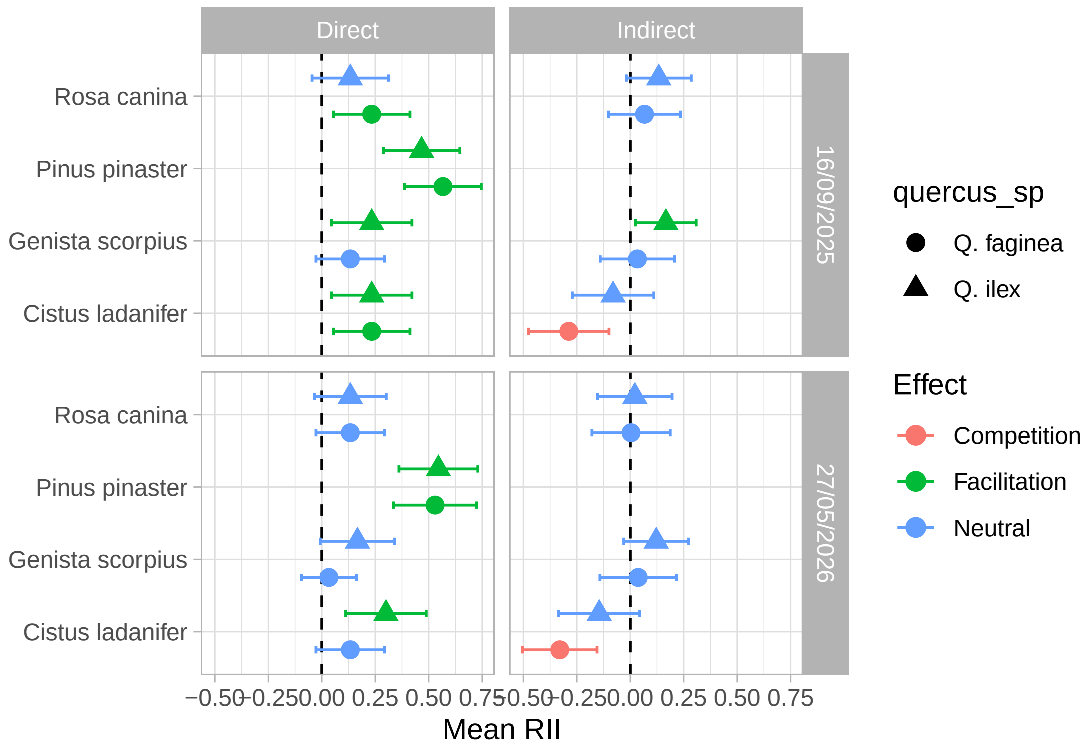
5.2 T-test and data normality

The analysis of the results using t-tests yields conclusions that are very similar to those obtained from the descriptive exploration of the data.

```
# Añadir nombre de especie
rii.qi <- aggID.df.rii.qi |>
  mutate(quercus_sp = "Q. ilex")

rii.qf <- aggID.df.rii.qf |>
  mutate(quercus_sp = "Q. faginea")

rii.all <- bind_rows(rii.qi, rii.qf) |>
  filter(surv_date != "12/06/2025")

# One-sample t-tests against RII = 0

ttest.rii <- rii.all |>
  group_by(
```

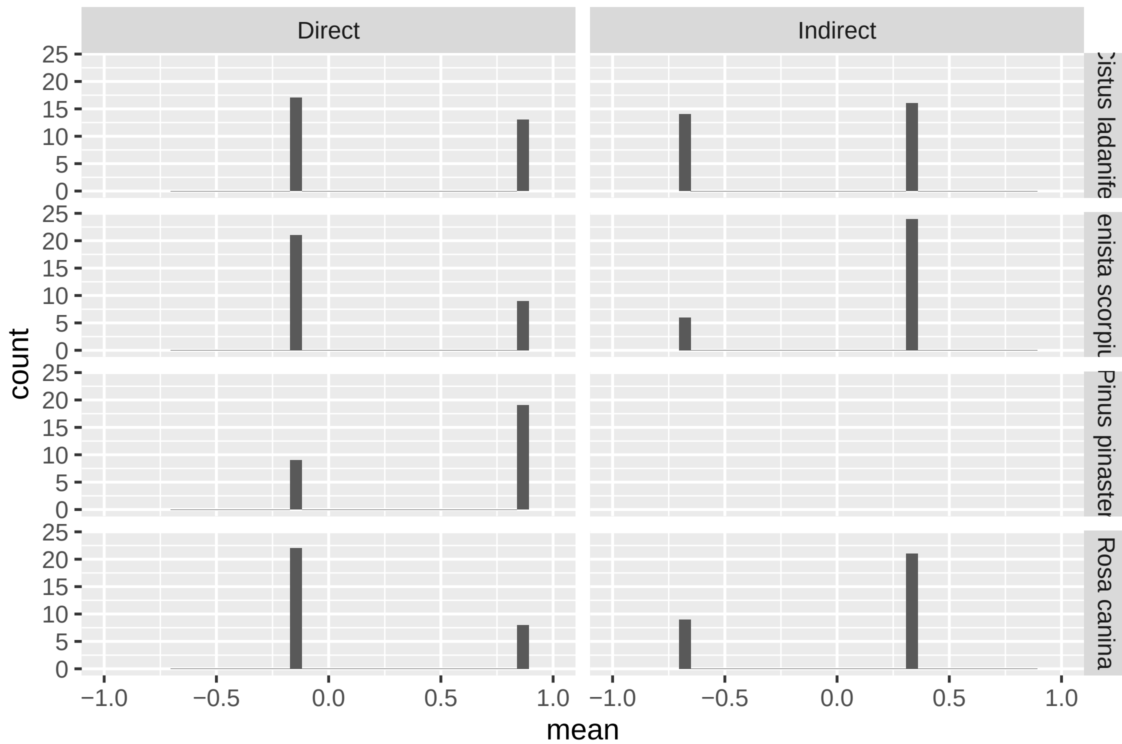
```r
    quercus_sp,
    surv_date,
    type_RII,
    Interacting_species
  ) |>
  summarise(
    test = list(t.test(mean, mu = 0)),
    .groups = "drop"
  ) |>
  mutate(tidy = map(test, broom::tidy)) |>
  unnest(tidy) |>
  select(
    quercus_sp,
    surv_date,
    type_RII,
    Interacting_species,
    estimate,
    statistic,
    parameter,
    conf.low,
    conf.high,
    p.value
  ) |>
  arrange(quercus_sp,
          surv_date,
          type_RII,
          Interacting_species)

ttest.rii <- ttest.rii |>
  mutate(
    interpretation = case_when(
      p.value < 0.05 & estimate > 0 ~ "Facilitation",
      p.value < 0.05 & estimate < 0 ~ "Competition",
      TRUE ~ "Neutral"
    )
  )


ggplot(
  ttest.rii,
  aes(
```

```
    x = estimate,
    y = Interacting_species,
    colour = interpretation
  )
) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_point(aes(shape = quercus_sp), size = 3,
             position = position_dodge(width = 1)) +
  geom_errorbarh(
    aes(
      xmin = conf.low,
      xmax = conf.high,
      shape = quercus_sp
    ),
    height = .2,
    position = position_dodge(width = 1)
  ) +
  facet_grid(
    surv_date ~  type_RII
  ) +
  theme_light() +
  labs(
    x = "Mean RII",
    y = "",
    colour = "Effect"
  )
```



However, the analysis of the data distribution reveals clear departures from normality. As shown in the following count histogram, for the May 2026 survival data, RII values collapse into only two numerical possibilities for each species and each RII type (direct/indirect).

```
ggplot(aggID.df.rii.qi |> filter(surv_date == "27/05/2026"),
       aes(x = mean)) +
  geom_histogram(bins = 30) +
  #geom_density() +
  coord_cartesian(xlim = c(-1, 1)) +
  facet_grid(Interacting_species~type_RII)
```



The corresponding table provides the exact numerical values and the number of individuals associated with each.

```
aggID.df.rii.qi |>
  filter(surv_date == "27/05/2026") |>
  group_by(Interacting_species, type_RII, mean) |>
  count(mean)
```

```
# A tibble: 15 x 4
# Groups:   Interacting_species, type_RII, mean [15]
   Interacting_species type_RII   mean     n
   <chr>               <chr>     <dbl> <int>
 1 Cistus ladanifer    Direct   -0.133    17
 2 Cistus ladanifer    Direct    0.867    13
 3 Cistus ladanifer    Indirect -0.679    14
 4 Cistus ladanifer    Indirect  0.321    16
 5 Genista scorpius    Direct   -0.133    21
 6 Genista scorpius    Direct    0.867     9
 7 Genista scorpius    Indirect -0.679     6
 8 Genista scorpius    Indirect  0.321    24
 9 Pinus pinaster      Direct   -0.133     9
10 Pinus pinaster      Direct    0.867    19
```

```
11 Pinus pinaster      Direct   NaN          2
12 Rosa canina         Direct    -0.133    22
13 Rosa canina         Direct     0.867     8
14 Rosa canina         Indirect  -0.679     9
15 Rosa canina         Indirect   0.321    21
```

A normal distribution of RII values was expected, given that the index is standardized between -1 and 1, and potentially skewed toward positive or negative values depending on species and interaction type.

An examination of the RII calculation formula may help explain this pattern (**?@eq-RII**). For binary variables, the expression simplifies exactly to:

$$RII_{ij} = S_{A_i} - S_{B_j}$$

{#eq-RII_summ}

because the four possible cases reproduce the same outcomes shown in table **?@tbl-posible.RII**:

S_{A_i}	S_{B_j}	RII_{ij}
1	0	1
0	1	-1
1	1	0
0	0	0

Since S_{A_i} is constant for each focal individual:

$$\overline{RII_i} = S_{A_i} - \bar{B}$$

{#eq-RII_summ2}

This has a direct consequence:

- if the focal individual survives, then $\overline{RII_i} = 1 - \bar{B}$,
- if the focal individual does not survive, then $\overline{RII_i} = 0 - \bar{B} = -\bar{B}$.

In other words, for each specific combination, the aggregated RII per individual can only take **two possible values**. In the following code example, it can be observed that the original calculation (**?@eq-RII**), the simplified formulation (**?@eq-RII_summ**), and the formulation where RII is expressed as the simple difference between survival in A and the mean of B (**?@eq-RII_summ2**) all yield identical results.

```
idA <- 1:5
A <- c(0,0,1,1,0)
```

```
B <- c(0,0,1,0,0)

ejemplo <- expand.grid(A = A, B = B) |>
  mutate(idA = rep(idA, 5),
    RII_orig = if_else(A + B == 0, 0, (A - B) / (A + B)),
         RII_summ = A - B)

media_B <- mean(ejemplo$B)
ejemplo |>
  mutate(media_B = media_B,
         A_media_B = A - media_B) |>
  group_by(idA) |>
  summarise(
    media_RII_orig = mean(RII_orig),
    media_RII_summ = mean(RII_summ),
    media_RII_summ2 = mean(A_media_B)
  )
```

```
# A tibble: 5 x 4
    idA media_RII_orig media_RII_summ media_RII_summ2
  <int>          <dbl>          <dbl>           <dbl>
1     1           -0.2           -0.2            -0.2
2     2           -0.2           -0.2            -0.2
3     3            0.8            0.8             0.8
4     4            0.8            0.8             0.8
5     5           -0.2           -0.2            -0.2
```

This would explain why:

- the distributions are non-normal,
- histograms show only a few discrete peaks,
- quantiles repeat across observations,
- the Shapiro–Wilk test rejects normality in all cases.

In practical terms, the mean RII per individual is equivalent to a **difference in survival probabilities**:

$$\overline{RII} = p_A - p_B$$

where:

- p_A is the probability of survival under the focal condition,
- p_B is the probability of survival under the reference condition.

References

Michalet, Richard, Shu-yan Chen, Li-zhe An, et al. 2014. "Communities: Are They Groups of Hidden Interactions?" *Journal of Vegetation Science* 26 (2): 207–18. https://doi.org/10.1111/jvs.12226.

Michalet, Richard, Jean-Paul Maalouf, and Patrick Al Hayek. 2017. "Direct Litter Interference and Indirect Soil Competitive Effects of Two Contrasting Phenotypes of a Spiny Legume Shrub Drive the Forb Composition of an Oromediterranean Community." *Oikos* 126 (8): 1090–100. https://doi.org/10.1111/oik.03800.

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